

```
In [37]: import pandas as pd  
df = pd.read_pickle("steam_cleaned.pkl")
```

Question 1: How has the average review score of Steam games changed over time, and does it differ by genre?

Insight

This visualization tracks the average review score of Steam games over time for the five most common genres. It highlights whether player satisfaction has changed over the years and shows if certain genres consistently receive higher review scores than others.

```
In [38]: import plotly.express as px

# Use existing release_year column
plot_df = df.dropna(
    subset=["release_year", "review_score", "primary_genre"]
).copy()

plot_df["release_year"] = plot_df["release_year"].astype(int)

# Top 5 genres
top_genres = (
    plot_df["primary_genre"]
    .value_counts()
    .head(5)
    .index
)

plot_df = plot_df[
    plot_df["primary_genre"].isin(top_genres)
]

# Average review score
avg_reviews = (
    plot_df
    .groupby(["release_year", "primary_genre"])["review_score"]
    .mean()
    .reset_index()
)

fig = px.line(
    avg_reviews,
    x="release_year",
    y="review_score",
    color="primary_genre",
    markers=True,
    color_discrete_map={
        "Action": "#0072B2",
        "Adventure": "#BDBDBD",
        "Casual": "#CFCFCF",
        "Indie": "#A6A6A6",
        "Simulation": "#D9D9D9",
    }
)

fig.update_traces(line=dict(width=3), marker=dict(size=5))

fig.update_layout(
    title="<b>Average review scores have remained consistently  
high across Steam genres over time</b>",
    template="simple_white",
    xaxis_title="Release Year",
    yaxis_title="Average Review Score",
```

```
        font=dict(size=14),
        legend_title="Genre"
    )

    fig.update_xaxes(
        dtick=1,
        showgrid=False
    )

    fig.update_yaxes(
        showgrid=False
    )

    # Annotation
    action = avg_reviews.query("primary_genre ==
    'Action']").sort_values("release_year")

    fig.add_annotation(
        x=action.iloc[-1]["release_year"],
        y=action.iloc[-1]["review_score"],
        text="Action remains consistently well-rated",
        showarrow=True,
        arrowhead=2
    )

    fig.show()
```

Conclusion

Average review scores across Steam's major genres declined during the platform's early growth period before gradually recovering in recent years. By 2025, Action and Adventure games consistently achieved some of the highest average review scores, suggesting that player satisfaction has improved despite the increasing number of game releases on Steam.

Question 2: Do higher-priced games receive better review scores, or is there little relationship between price and quality?

Insight

This visualization examines whether there is a relationship between a game's price and its average review score. By displaying thousands of individual games, it helps determine whether paying more generally results in better-rated games or whether review quality is largely independent of price.

```
In
[39]: import pandas as pd
import plotly.express as px

plot_df = df.dropna(subset=["price", "review_score"]).copy()

# Create price categories only for this visualization
plot_df["price_range"] = pd.cut(
    plot_df["price"],
    bins=[-0.01, 0, 5, 10, 20, 30, plot_df["price"].max()],
    labels=[
        "Free",
        "€0-5",
        "€5-10",
        "€10-20",
        "€20-30",
        "€30+"
    ]
)

fig = px.box(
    plot_df,
    x="price_range",
    y="review_score",
    points=False,
    color_discrete_sequence=["#0072B2"]
)

fig.update_layout(
    title="<b>Higher prices do not consistently lead to better  
review scores</b>",
    template="simple_white",
    xaxis_title="Price Range",
    yaxis_title="Review Score",
    showlegend=False,
    font=dict(size=14)
)

fig.update_xaxes(
    showgrid=False,
    categoryorder="array",
    categoryarray=[
        "Free",
        "€0-5",
        "€5-10",
        "€10-20",
        "€20-30",
        "€30+"
    ]
)

fig.update_yaxes(showgrid=False)
```

```
fig.add_annotation(  
    x=5,  
    y=96,  
    text="Median review scores remain similar across price  
ranges",  
    showarrow=False,  
    bgcolor="white",  
    font=dict(size=12)  
)  
  
fig.show()
```

Conclusion

The review score distributions are remarkably similar across most price ranges, with median scores consistently around 80–85. While games priced above €30 exhibit slightly greater variability and a lower median review score, the overall pattern suggests that higher prices do not consistently result in better player ratings. Quality appears to be largely independent of price, with highly rated games available across all price categories.

Question 3

Which genres consistently receive the highest review scores while also attracting large numbers of reviews?

Insight

This visualization compares Steam genres based on their average review score and average number of reviews. Genres positioned in the upper-right corner combine both strong player satisfaction and high community engagement, indicating the most successful categories on the platform.

```
In [40]: import plotly.express as px

# Keep required columns
plot_df = df.dropna(
    subset=["primary_genre", "review_score", "review_count"]
).copy()

# Genre statistics
genre_stats = (
    plot_df
    .groupby("primary_genre")
    .agg(
        avg_review_score=("review_score", "mean"),
        avg_review_count=("review_count", "mean"),
        games=("primary_genre", "count")
    )
    .reset_index()
)

# Keep only genres with enough games
genre_stats = genre_stats[
    genre_stats["games"] >= 100
]

# Keep the 10 largest genres
genre_stats = genre_stats.nlargest(10, "games")

# Highlight the genre with the highest review score
highlight = genre_stats.loc[
    genre_stats["avg_review_score"].idxmax(),
    "primary_genre"
]

genre_stats["Highlight"] = genre_stats["primary_genre"].apply(
    lambda x: highlight if x == highlight else "Other Genres"
)

fig = px.scatter(
    genre_stats,
    x="avg_review_count",
    y="avg_review_score",
    size="games",
    text="primary_genre",
    color="Highlight",
    size_max=45,
    color_discrete_map={
        highlight: "#0072B2",
        "Other Genres": "#BDBDBD"
    }
)

fig.update_traces(
```

```
        textposition="top center",
        marker=dict(
            line=dict(color="white", width=1),
            opacity=0.9
        )
    )

fig.update_layout(
    title="<b>Adventure and Action combine high player ratings  
with strong engagement</b>",
    template="simple_white",
    xaxis_title="Average Number of Reviews",
    yaxis_title="Average Review Score",
    showlegend=False,
    font=dict(size=14)
)

fig.update_xaxes(showgrid=False)
fig.update_yaxes(showgrid=False)

fig.add_annotation(
    x=3500,
    y=80,
    text="Adventure receives the<br>highest average rating",
    showarrow=False,
    bgcolor="white",
    bordercolor="lightgray",
    borderwidth=1
)

fig.show()
```

Conclusion

Adventure and Action stand out by achieving both high average review scores and large numbers of player reviews, indicating a balance between quality and popularity. Although genres such as Casual and Indie also receive strong ratings, they attract fewer reviews on average, suggesting that highly rated games do not always generate the largest player communities.

Question 4: How has the number of games released each year changed across different genres?

Insight

This visualization tracks the yearly number of game releases for Steam's five most common genres. It highlights which genres have contributed most to Steam's rapid expansion and reveals how the platform's genre landscape has evolved over the past two decades.

```
In [41]: import plotly.express as px

plot_df = df.dropna(subset=["release_year",
                           "primary_genre"]).copy()

plot_df["release_year"] = plot_df["release_year"].astype(int)

# Top 5 genres by number of games
top_genres = (
    plot_df["primary_genre"]
    .value_counts()
    .head(5)
    .index
)

plot_df = plot_df[
    plot_df["primary_genre"].isin(top_genres)
]

yearly = (
    plot_df
    .groupby(["release_year", "primary_genre"])
    .size()
    .reset_index(name="Games Released")
)

fig = px.line(
    yearly,
    x="release_year",
    y="Games Released",
    color="primary_genre",
    color_discrete_map={
        "Action": "#0072B2",
        "Adventure": "#BDBDBD",
        "Casual": "#C9C9C9",
        "Indie": "#AFAFAF",
        "Simulation": "#D9D9D9"
    },
    markers=True
)

fig.update_traces(line=dict(width=3), marker=dict(size=5))

fig.update_layout(
    title="<b>Steam's rapid growth has been driven primarily by  
Action games since 2015</b>",
    template="simple_white",
    xaxis_title="Release Year",
    yaxis_title="Number of Games Released",
    legend_title="Genre",
    font=dict(size=14)
)
```

```
fig.update_xaxes(
    showgrid=False,
    dtick=2
)

fig.update_yaxes(
    showgrid=False
)

# Latest Action value
action = yearly[
    yearly["primary_genre"] == "Action"
].sort_values("release_year")

fig.add_annotation(
    x=action.iloc[-1]["release_year"],
    y=action.iloc[-1]["Games Released"],
    text="Action dominates<br>recent releases",
    showarrow=True,
    arrowhead=2
)

fig.show()
```

Conclusion

Steam experienced rapid growth in annual game releases after 2014, with Action becoming the dominant genre and contributing the largest share of new titles. Although all major genres expanded over time, the sharp decline observed in 2025 is likely due to the dataset containing only partial-year release data rather than indicating a genuine decrease in game production.

Question 5: Which developers have released the most highly rated games (minimum number of games to avoid one-hit wonders)?

Insight

This visualization compares developers based on their average review score while filtering out studios with very small game catalogues. It highlights established developers that consistently produce highly rated games rather than one-hit successes.

```
In [42]: import plotly.express as px

plot_df = df.dropna(
    subset=["developer", "review_score"]
).copy()

developer_stats = (
    plot_df
    .groupby("developer")
    .agg(
        average_review_score=("review_score", "mean"),
        games=("developer", "count")
    )
    .reset_index()
)

# Keep developers with at least 10 released games
developer_stats = developer_stats[
    developer_stats["games"] >= 10
]

# Top 10 highest-rated developers
developer_stats = (
    developer_stats
    .sort_values("average_review_score", ascending=False)
    .head(10)
)

fig = px.bar(
    developer_stats,
    x="average_review_score",
    y="developer",
    orientation="h",
    text="average_review_score",
    color=[
        "#0072B2" if i == 0 else "#BDBDBD"
        for i in range(len(developer_stats))
    ]
)

fig.update_traces(
    texttemplate="%{text:.1f}",
    textposition="outside"
)

fig.update_layout(
    title="<b>Several established developers consistently achieve  
outstanding player ratings</b>",
    template="simple_white",
    xaxis_title="Average Review Score",
    yaxis_title="Developer",
    showlegend=False,
```

```
        font=dict(size=14)
    )

    fig.update_xaxes(showgrid=False)
    fig.update_yaxes(showgrid=False)

    fig.add_annotation(
        x=92,
        y=0.6,
        text="Highest average rating<br>(minimum 10 games)",
        showarrow=True,
        arrowhead=2,
        ax=-50,
        ay=-35,
        bgcolor="white",
        bordercolor="lightgray",
        borderwidth=1
    )
    fig.show()
```

Conclusion

Among developers with at least 10 released games, Blam! Games achieves the highest average review score, demonstrating consistently strong player satisfaction across its portfolio. The relatively small differences among the top-ranked developers suggest that several established studios are capable of maintaining excellent game quality over multiple releases.

Question 6: Do free games receive more reviews than paid games, and how do their review scores compare?

Insight

This visualization compares free and paid Steam games based on their average number of reviews and average review scores. It highlights whether free-to-play titles generate greater player engagement while maintaining review scores comparable to paid games.

```
In
[43]: import plotly.express as px
import pandas as pd

plot_df = df.dropna(subset=["price", "review_count",
"review_score"]).copy()

plot_df["Game Type"] = plot_df["price"].apply(
    lambda x: "Free" if x == 0 else "Paid"
)

summary = (
    plot_df
    .groupby("Game Type")
    .agg(
        average_reviews=("review_count", "mean"),
        average_score=("review_score", "mean"),
        games=("Game Type", "count")
    )
    .reset_index()
)

fig = px.scatter(
    summary,
    x="average_reviews",
    y="average_score",
    size="games",
    text="Game Type",
    color="Game Type",
    color_discrete_map={
        "Free": "#0072B2",
        "Paid": "#BDBDBD"
    },
    size_max=80
)

fig.update_traces(
    textposition="top center",
    marker=dict(
        line=dict(color="white", width=2)
    )
)

fig.update_layout(
    title="<b>Free games attract substantially more reviews while
maintaining comparable player ratings</b>",
    template="simple_white",
    xaxis_title="Average Number of Reviews",
    yaxis_title="Average Review Score",
    legend_title="",
    font=dict(size=14)
)
```

```
fig.update_xaxes(showgrid=False)
fig.update_yaxes(showgrid=False)

highest = summary.loc[summary["average_reviews"].idxmax()]

fig.add_annotation(
    x=highest["average_reviews"],
    y=highest["average_score"],
    text="Free games generate<br>greater player engagement",
    showarrow=True,
    arrowhead=2,
    ax=40,
    ay=-40,
    bgcolor="white",
    bordercolor="lightgray"
)

fig.show()
```

Conclusion

Free-to-play games attract substantially more player reviews than paid titles, indicating significantly higher levels of player engagement. Despite this difference in popularity, both free and paid games achieve very similar average review scores, suggesting that pricing has little influence on overall player satisfaction.

Question 7: How are review scores distributed across the Steam catalogue, and what does this reveal about overall game quality?

Insight

This histogram shows the distribution of review scores across all Steam games. It reveals whether most games receive positive reviews or whether ratings are evenly spread across the review score scale.

```
In [44]: import plotly.express as px

plot_df = df.dropna(subset=["review_score"]).copy()

fig = px.histogram(
    plot_df,
    x="review_score",
    nbins=20,
    color_discrete_sequence=["#0072B2"]
)

fig.update_layout(
    title="<b>Most Steam games receive positive review scores,  
with ratings concentrated above 70</b>",
    template="simple_white",
    xaxis_title="Review Score",
    yaxis_title="Number of Games",
    bargap=0.05,
    font=dict(size=14),
    showlegend=False
)

fig.update_xaxes(
    showgrid=False,
    range=[0, 100]
)

fig.update_yaxes(showgrid=False)

fig.add_annotation(
    x=82,
    y=plot_df["review_score"].value_counts().max()*0.85,
    text="Most games receive<br>positive player reviews",
    showarrow=True,
    arrowhead=2,
    ax=-60,
    ay=-40,
    bgcolor="white",
    bordercolor="lightgray",
    borderwidth=1
)

fig.show()
```

Conclusion

The distribution of review scores is heavily concentrated between 70 and 90, indicating that most Steam games receive generally positive player feedback. Relatively few games receive very low review scores, suggesting that the overall quality of games available on Steam is perceived favorably by players.

Question 8: What pricing strategy dominates the Steam marketplace, and how are games distributed across different price points?

Insight

This histogram shows the distribution of Steam game prices. It highlights the most common price ranges and reveals whether Steam's marketplace is dominated by low-cost, premium-priced, or free games.

```
In [45]: import plotly.express as px

plot_df = df.dropna(subset=["price"]).copy()

# Remove extreme price outliers for readability
plot_df = plot_df[
    plot_df["price"] <= plot_df["price"].quantile(0.99)
]

fig = px.histogram(
    plot_df,
    x="price",
    nbins=25,
    color_discrete_sequence=["#0072B2"]
)

fig.update_layout(
    title="<b>Most Steam games are priced below €10, with  
relatively few premium-priced titles</b>",
    template="simple_white",
    xaxis_title="Price (€)",
    yaxis_title="Number of Games",
    bargap=0.05,
    font=dict(size=14),
    showlegend=False
)

fig.update_xaxes(
    showgrid=False
)

fig.update_yaxes(
    showgrid=False
)

fig.add_annotation(
    x=6,
    y=plot_df["price"].value_counts().max()*0.8,
    text="Most games fall<br>below €10",
    showarrow=True,
    arrowhead=2,
    ax=50,
    ay=-40,
    bgcolor="white",
    bordercolor="lightgray",
    borderwidth=1
)

fig.show()
```

Conclusion

The distribution of Steam game prices is heavily skewed toward lower price points, with most games costing less than €10. Premium-priced titles are comparatively rare, indicating that Steam's marketplace is largely dominated by affordable games, making the platform accessible to a broad range of players.

Question 9: Which Steam titles have achieved the greatest levels of player engagement based on review volume?

Insight

This visualization ranks the Steam games with the highest number of player reviews. A large number of reviews reflects sustained popularity, an active player base, and long-term community engagement.

```
In [46]: import plotly.express as px

plot_df = df.dropna(subset=["name", "review_count"]).copy()

# Top 15 most reviewed games
top_games = (
    plot_df
    .sort_values("review_count", ascending=False)
    .head(15)
)

# Highlight the most-reviewed game
top_games["Highlight"] = [
    "Most Reviewed" if i == 0 else "Other Games"
    for i in range(len(top_games))
]

fig = px.bar(
    top_games,
    x="review_count",
    y="name",
    orientation="h",
    color="Highlight",
    text="review_count",
    color_discrete_map={
        "Most Reviewed": "#0072B2",
        "Other Games": "#BDBDBD"
    }
)

fig.update_traces(
    texttemplate="%{text:,.0f}",
    textposition="outside"
)

fig.update_layout(
    title="<b>Counter-Strike 2 leads Steam with the largest player  
review community</b>",
    template="simple_white",
    xaxis_title="Number of Reviews",
    yaxis_title="Game",
    showlegend=False,
    font=dict(size=14)
)

fig.update_xaxes(showgrid=False)

fig.update_yaxes(
    showgrid=False,
    autorange="reversed"
)
```

```
highest = top_games.iloc[0]

fig.add_annotation(
    x=highest["review_count"],
    y=highest["name"],
    text="Largest review community",
    showarrow=True,
    arrowhead=2,
    ax=-70,
    ay=-30,
    bgcolor="white",
    bordercolor="lightgray",
    borderwidth=1
)

fig.show()
```

Conclusion

The most-reviewed Steam games have accumulated millions of player reviews, reflecting exceptionally large and active player communities. Counter-Strike 2 stands far ahead of all other titles, while games such as PUBG, Dota 2, and Grand Theft Auto V also demonstrate long-term popularity through sustained player engagement and continuous community activity

Question 10: What relationship exists between player engagement and player satisfaction on Steam?

Insight

This heatmap illustrates the relationship between player engagement and game quality by comparing review counts with review scores. Areas with darker colours indicate where the largest number of Steam games are concentrated, making overall patterns easier to identify.

```
In [47]: import plotly.express as px
import numpy as np

plot_df = df.dropna(
    subset=["review_count", "review_score"]
).copy()

# Log-transform review count for readability
plot_df["log_reviews"] = np.log10(plot_df["review_count"] + 1)

fig = px.density_heatmap(
    plot_df,
    x="log_reviews",
    y="review_score",
    nbinsx=35,
    nbinsy=25,
    color_continuous_scale="Turbo"
)

fig.update_layout(
    title="<b>Highly reviewed games are generally well-rated,  
although quality varies across all popularity levels</b>",
    template="simple_white",
    xaxis_title="Log10(Number of Reviews)",
    yaxis_title="Review Score",
    coloraxis_colorbar_title="Number<br>of Games",
    font=dict(size=14),
    coloraxis_colorbar=dict(
        thickness=18,
        len=0.75
    )
)

fig.update_xaxes(showgrid=False)
fig.update_yaxes(showgrid=False)

fig.add_annotation(
    x=3.8,
    y=84,
    text="Most games cluster<br>between 70-90 review score",
    showarrow=True,
    arrowhead=2,
    ax=-50,
    ay=-60,
    bgcolor="white",
    bordercolor="lightgray",
    borderwidth=1
)
fig.show()
```

Conclusion

The highest concentration of Steam games falls within review scores of approximately 70–90, regardless of the number of player reviews received. While games with very large review counts are generally well-rated, the heatmap indicates that strong player ratings are achieved across a wide range of popularity levels, suggesting that review count alone is not a reliable indicator of game quality.

Question 11: How concentrated is game production among Steam developers, and which studios contribute the largest share of releases?

Insight

This visualization ranks the most prolific developers on Steam based on the total number of games released. It highlights the studios that have contributed the largest number of titles to the platform, regardless of review scores or popularity.

```
In
[48]: import plotly.express as px
import plotly.graph_objects as go

plot_df = df.dropna(subset=["developer"]).copy()

developer_count = (
    plot_df["developer"]
    .value_counts()
    .head(15)
    .reset_index()
)

developer_count.columns = ["Developer", "Games Released"]

fig = go.Figure()

# Lollipop stems
fig.add_trace(
    go.Scatter(
        x=developer_count["Games Released"],
        y=developer_count["Developer"],
        mode="lines",
        line=dict(color="#C7C7C7", width=2),
        showlegend=False
    )
)

# Lollipop heads
fig.add_trace(
    go.Scatter(
        x=developer_count["Games Released"],
        y=developer_count["Developer"],
        mode="markers+text",
        marker=dict(
            size=12,
            color=[
                "#0072B2" if i == 0 else "#BDBDBD"
                for i in range(len(developer_count))
            ],
            line=dict(color="white", width=1)
        ),
        text=developer_count["Games Released"],
        textposition="middle right",
        showlegend=False
    )
)

# Draw horizontal stems
for i in range(len(developer_count)):
    fig.add_shape(
        type="line",
        x0=0,
```

```
        x1=developer_count.iloc[i]["Games Released"],
        y0=i,
        y1=i,
        line=dict(color="#D9D9D9", width=2)
    )

fig.update_layout(
    title="<b>Several developers have built exceptionally large
game portfolios on Steam</b>",
    template="simple_white",
    xaxis_title="Number of Games Released",
    yaxis_title="Developer",
    font=dict(size=14),
    height=700
)

fig.update_xaxes(showgrid=False)
fig.update_yaxes(
    showgrid=False,
    autorange="reversed"
)

highest = developer_count.iloc[0]

fig.add_annotation(
    x=highest["Games Released"],
    y=highest["Developer"],
    text="Largest game catalogue",
    showarrow=True,
    arrowhead=2,
    ax=-60,
    ay=-40,
    bgcolor="white",
    bordercolor="lightgray",
    borderwidth=1
)

fig.show()
```

Conclusion

A relatively small number of developers have released exceptionally large numbers of games on Steam, with Choice of Games leading the platform by a considerable margin. The steep decline after the top few developers suggests that most studios focus on producing a much smaller portfolio of titles rather than maximizing release volume.

Question 12: How concentrated is Steam's catalogue across major game genres, and which genres dominate the platform?

Insight

This treemap illustrates how Steam's catalogue is distributed across its major game genres. Larger rectangles represent genres with more released games, providing a clear overview of the platform's composition.

```
In [49]: import plotly.express as px

plot_df = df.dropna(subset=["primary_genre"]).copy()

genre_counts = (
    plot_df["primary_genre"]
    .value_counts()
    .reset_index()
)

genre_counts.columns = ["Genre", "Games"]

fig = px.treemap(
    genre_counts,
    path=["Genre"],
    values="Games",
    color="Games",
    color_continuous_scale="Blues"
)

fig.update_traces(
    textinfo="label+value+percent root"
)

fig.update_layout(
    title="<b>Action and Adventure account for the largest share  
of Steam's game catalogue</b>",
    template="simple_white",
    font=dict(size=14),
    coloraxis_showscale=False,
    margin=dict(t=70, l=20, r=20, b=20)
)

fig.show()
```

Conclusion

Steam's catalogue is dominated by Action and Adventure games, which together account for the largest share of releases on the platform. Although genres such as Casual, Indie, and Simulation also contribute substantially to the catalogue, the treemap highlights an uneven distribution in which a small number of genres represent the majority of available titles.

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